

Merrivoss Bioanalysis Inc, a Tarnovell company

TP04 inflammatory kinase follow-up: 40-compound cardiac and CYP screen

Final Report

Study MV-2024-358 (LIMS M24/BIO/08257)

Client: Tervemere Pharma AG

Study Director: Dr P. Nandekuri

Report date: December 19, 2024

1 Summary

40 compounds from the Tervemere aminopyrazolopyrimidine series 04 were tested for Tervemere Pharma AG in the IKr inhibition, Five-isoform DDI screen, CYP3A4 inhibition curves. Testing started on December 5, 2024 and results were reported on December 19, 2024. All runs met matrix acceptance criteria. No compound inhibited hERG with an IC50 below 1 µM. Most compounds were weak or inactive (IC50 at or above 10 µM, or above the top concentration). Inhibition above 50% at 10 µM was most frequent for CYP3A4.

Table 1 Study summary

Metric	Value
Project	MV-2024-358
Client	Tervemere Pharma AG
Compounds tested	40
Series / chemotype	Tervemere aminopyrazolopyrimidine series 04 (Aminopyrazolopyrimidine)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 µM, CYP3A4 inhibition IC50
Turnaround (calendar days)	14
Price	USD 24,700
hERG IC50 <1 µM	0
hERG IC50 1-10 µM	9
hERG IC50 ≥10 µM	31
hERG median IC50 (µM)	16
E-4031 control IC50 (nM)	15
CYP1A2 >50% inhibition @10 µM	5
CYP2C9 >50% inhibition @10 µM	6
CYP2C19 >50% inhibition @10 µM	7
CYP2D6 >50% inhibition @10 µM	5
CYP3A4 >50% inhibition @10 µM	9
Compounds with any CYP >50%	12
Ketoconazole (3A4) IC50 (µM)	0.022

2 Objective

To rank the test article set for IKr inhibition, Five-isoform DDI screen, CYP3A4 inhibition curves liability so that Tervemere Pharma AG could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one accession lot on December 5, 2024 and logged against M24/BIO/08257. Identity and concentration were taken as supplied by the client.

4 Methods

IKr inhibition

CHO cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells were held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final), with 3-5 minutes per concentration.

- Matrix acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current amplitude ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- Positive control E-4031 on every run; IC₅₀ must fall within 5-50 nM
- IC₅₀ from a four-parameter logistic fit of mean tail-current inhibition; $n \geq 2$ cells per compound

Five-isoform DDI screen

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation was measured by LC-MS/MS against vehicle (0.1% DMSO) incubations.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Results are reported as mean percent inhibition relative to vehicle. Values above 50% at 10 μ M are flagged for follow-up; small negative values indicate no inhibition.

CYP3A4 inhibition curves

Compounds were run in a 7-point concentration response (top concentration 50 μ M, 3-fold dilutions) against CYP3A4 with midazolam or testosterone as probe; ketoconazole was the positive control.

5 Results

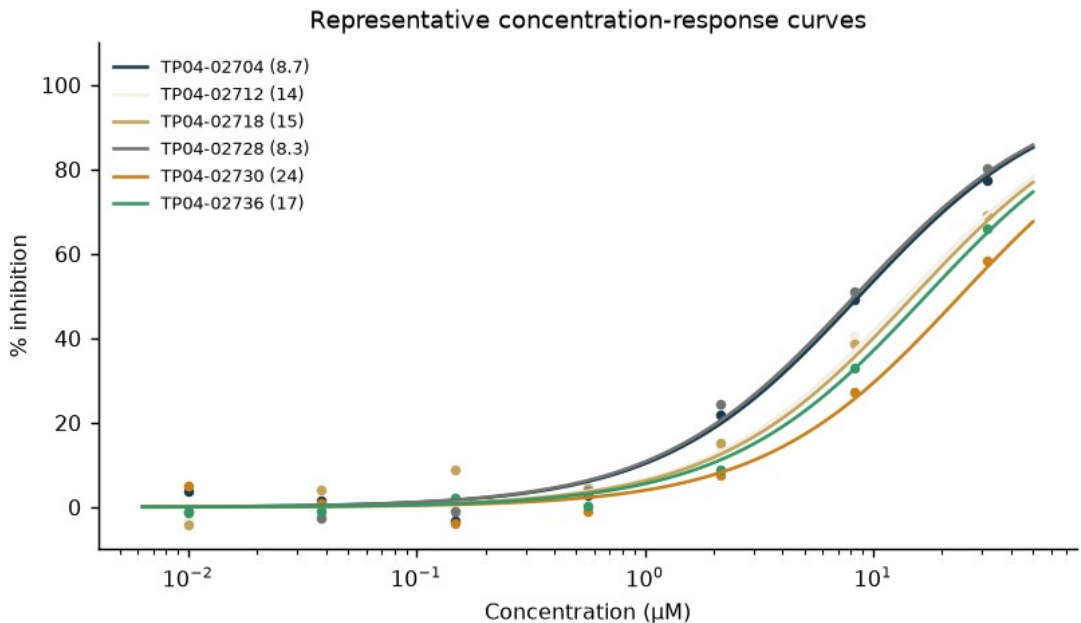
hERG results

All 40 compounds were tested. No compound inhibited hERG with an IC₅₀ below 1 μ M. Most compounds were weak or inactive (IC₅₀ at or above 10 μ M, or above the top concentration). The median IC₅₀ was 16 μ M (0 below 1 μ M, 9 between 1 and 10 μ M, 31 at or above 10 μ M). E-4031 gave an IC₅₀ of 15 nM, within the acceptance range.

Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ rep 1 (μ M)	IC ₅₀ rep 2 (μ M)	IC ₅₀ rep 3 (μ M)	IC ₅₀ mean (μ M)	% inhibition @10 μ M	n	QC note
TP04-02704	8.7	6.6		7.6	59	2	
TP04-02712	14	9.8		12	40	2	
TP04-02718	15	14		14	40	2	
TP04-02724	>30	>30		>30	25	2	
TP04-02728	8.3	7.2		7.7	59	2	
TP04-02730	24	28		26	32	2	
TP04-02736	17	13	16	15	42	3	
TP04-02740	15	15	9.8	13	35	3	
TP04-02748	16	16		16	41	2	
TP04-02752	14	12		13	35	2	
TP04-02764	10	5.7		7.7	53	2	
TP04-02766	11	16	28	17	39	3	
TP04-02770	>30	>30		>30	27	2	
TP04-02772	5.4	4.4		4.9	67	2	
TP04-02776	24	21	>30	23	33	3	
TP04-02782	29	>30		>30	18	2	
TP04-02784	6.8	14	13	11	47	3	
TP04-02794	7.9	6.7		7.3	55	2	
TP04-02800	17	>30		>30	27	2	
TP04-02806	>30	>30		>30	20	2	
TP04-02812	18	10		14	40	2	
TP04-02814	16	16		16	37	2	
TP04-02820	26	22		24	32	2	
TP04-02824	7.6	4.2		5.6	66	2	
TP04-02826	>30	>30		>30	29	2	
TP04-02830	6.3	6.4		6.4	56	2	
TP04-02832	19	20		19	41	2	
TP04-02842	22	29		25	28	2	
TP04-02844	11	8.1		9.5	54	2	
TP04-02848	6.7	7.7		7.1	54	2	
TP04-02850	16	19		17	40	2	
TP04-02856	14	13		14	43	2	
TP04-02860	11	9.3		10	50	2	
TP04-02862	>30	>30	26	>30	31	3	
TP04-02866	>30	>30		>30	25	2	
TP04-02868	28	22		25	30	2	
TP04-02878	23	25	18	22	29	3	
TP04-02884	>30	>30		>30	21	2	
TP04-02886	>30	>30		>30	15	2	

TP04-02890	21	>30		>30	30	2	
Top concentration 30 µM; 6-point, 3-fold dilution; positive control E-4031 IC50 15 nM (acceptance 5-50 nM)							



CYP inhibition results

Inhibition above 50% at 10 µM was most frequent for CYP3A4. In total 12 compounds inhibited at least one isoform by more than 50% at 10 µM (CYP3A4: 9; CYP2D6: 5; CYP2C19: 7; CYP2C9: 6; CYP1A2: 5). Ketoconazole inhibited CYP3A4 with an IC50 of 0.022 µM.

Table 3 Percent inhibition at 10 µM

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
TP04-02704	25	0	8	2	21
TP04-02712	1	12	5	3	3
TP04-02718	3	7	0	2	4
TP04-02724	42	6	8	51	47
TP04-02728	37	91	100	63	85
TP04-02730	-1	-3	9	4	-2
TP04-02736	4	3	4	2	9
TP04-02740	-6	3	-5	15	6
TP04-02748	2	1	11	1	3
TP04-02752	17	86	39	5	56
TP04-02764	57	21	58	95	81
TP04-02766	9	11	3	-8	11
TP04-02770	-2	7	-1	1	28
TP04-02772	72	17	11	4	6
TP04-02776	21	1	13	4	-2
TP04-02782	-5	-1	0	10	40
TP04-02784	-8	6	-1	-0	-6
TP04-02794	7	8	6	8	4
TP04-02800	90	55	88	100	97
TP04-02806	10	4	-1	1	-0
TP04-02812	3	17	-0	-11	-0

TP04-02814	6	2	-2	-0	9
TP04-02820	6	3	-0	5	22
TP04-02824	6	7	-2	-6	9
TP04-02826	4	11	-0	3	-2
TP04-02830	21	95	77	24	91
TP04-02832	31	97	83	30	98
TP04-02842	85	37	65	90	81
TP04-02844	5	16	2	6	-2
TP04-02848	75	3	25	6	12
TP04-02850	-1	4	1	5	9
TP04-02856	3	6	-0	-2	-6
TP04-02860	24	96	67	34	80
TP04-02862	-0	-3	0	-5	7
TP04-02866	3	-7	1	-7	2
TP04-02868	1	4	5	2	12
TP04-02878	11	37	36	2	75
TP04-02884	8	15	14	0	6
TP04-02886	-2	5	8	-12	-3
TP04-02890	7	-5	2	4	11

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.38; ticlopidine (CYP2C19) 1.4; quinidine (CYP2D6) 0.041; ketoconazole (CYP3A4) 0.022

CYP3A4 concentration response

Table 4 CYP3A4 IC₅₀ (μ M)

Compound	IC ₅₀ (μ M)	Probe substrate
TP04-02704	36	midazolam
TP04-02712	>50	midazolam
TP04-02718	>50	testosterone
TP04-02724	11	midazolam
TP04-02728	1.9	midazolam
TP04-02730	>50	midazolam
TP04-02736	>50	testosterone
TP04-02740	>50	midazolam
TP04-02748	>50	testosterone
TP04-02752	5.6	midazolam
TP04-02764	2.5	midazolam
TP04-02766	>50	midazolam
TP04-02770	29	testosterone
TP04-02772	>50	testosterone
TP04-02776	>50	midazolam
TP04-02782	18	testosterone
TP04-02784	>50	testosterone
TP04-02794	>50	midazolam
TP04-02800	0.18	midazolam
TP04-02806	>50	midazolam
TP04-02812	>50	midazolam
TP04-02814	>50	testosterone
TP04-02820	39	testosterone
TP04-02824	>50	testosterone
TP04-02826	>50	testosterone
TP04-02830	0.68	midazolam
TP04-02832	0.17	testosterone
TP04-02842	0.99	midazolam
TP04-02844	>50	midazolam

TP04-02848	>50	testosterone
TP04-02850	>50	midazolam
TP04-02856	>50	midazolam
TP04-02860	3.5	midazolam
TP04-02862	>50	midazolam
TP04-02866	>50	testosterone
TP04-02868	>50	midazolam
TP04-02878	2.7	midazolam
TP04-02884	>50	midazolam
TP04-02886	>50	midazolam
TP04-02890	>50	testosterone

7-point curve, top 50 μ M; ketoconazole control

6 Discussion

No compound inhibited hERG with an IC₅₀ below 1 μ M. Most compounds were weak or inactive (IC₅₀ at or above 10 μ M, or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4. Inhibition values are in vitro, at a fixed concentration, and do not by themselves predict clinical interaction risk; free exposure estimates are needed for that assessment.

7 Conclusion

The data support ranking within the set. Recommended next step: carry the compounds without CYP flags into exposure studies and run CYP3A4 IC₅₀s for flagged analogues before in vivo work.

Name	Role	Signature	Date
Priya Nandekuri	Study Director	<i>Nandekuri</i>	December 19, 2024
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